

Hayden Fuller

Modern Communication Systems F25

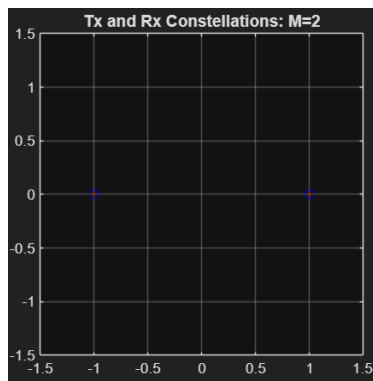
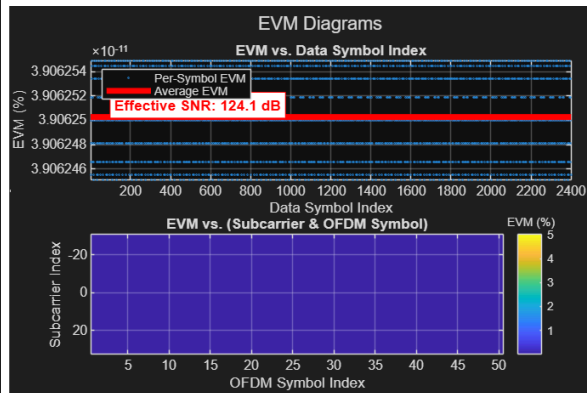
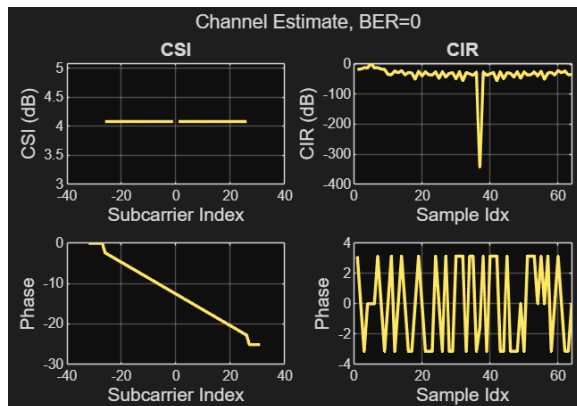
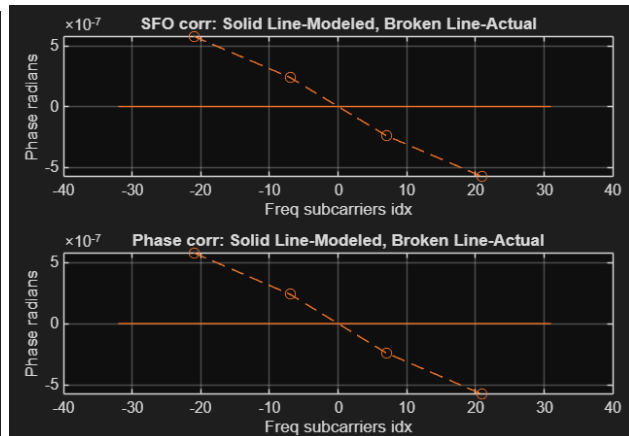
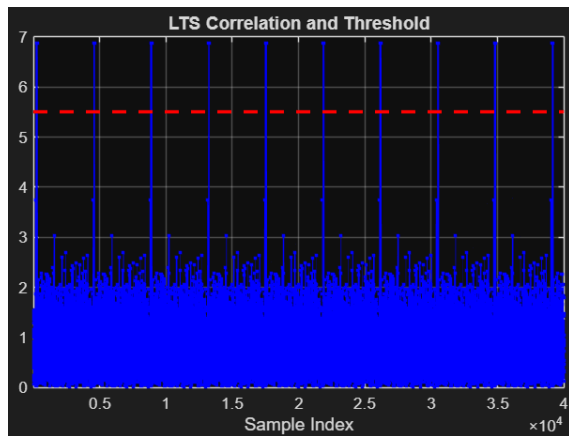
Lab 4

10/27/2025

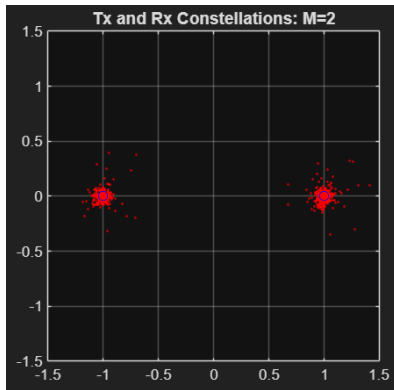
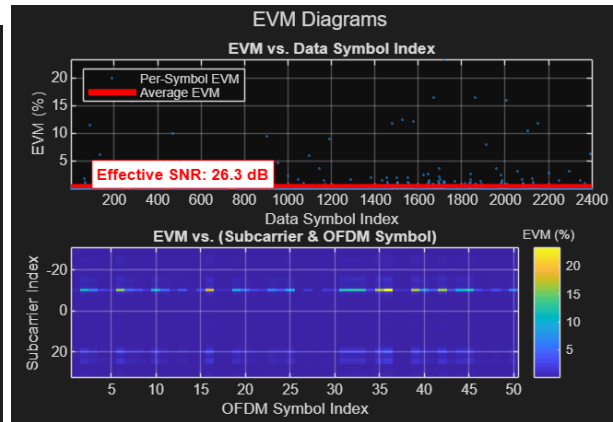
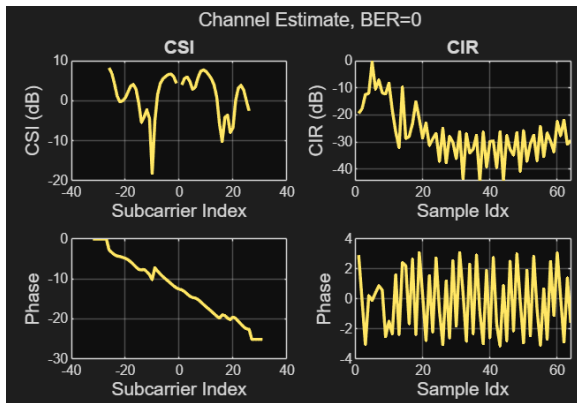
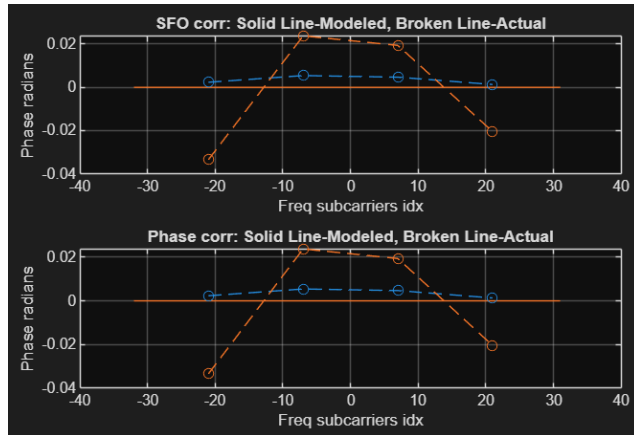
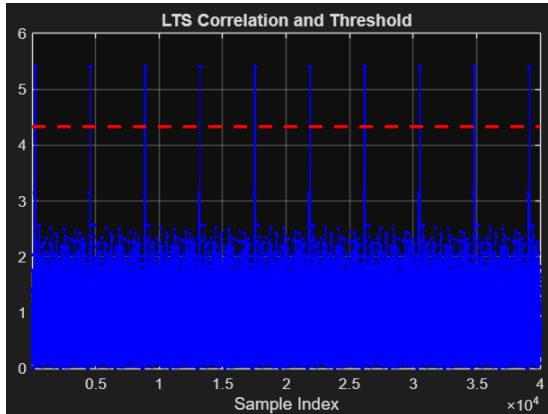
Matlab Simulations:

Implementation results/graphs

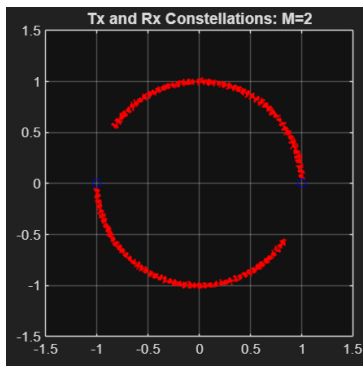
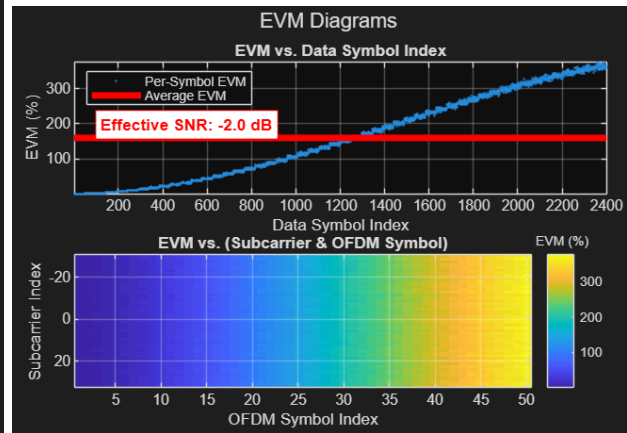
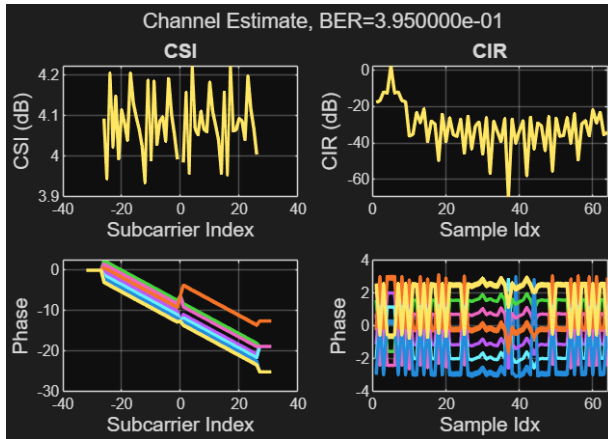
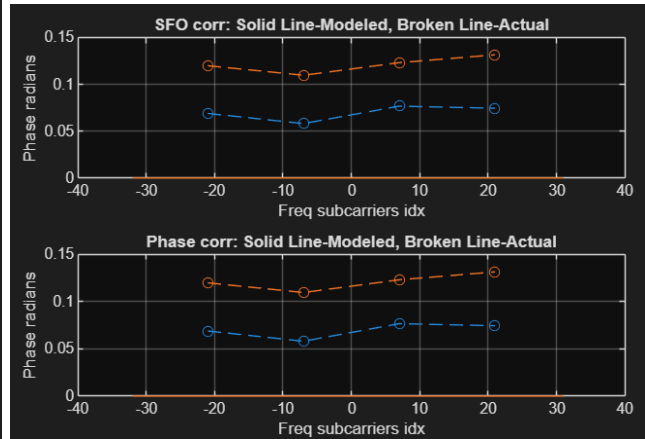
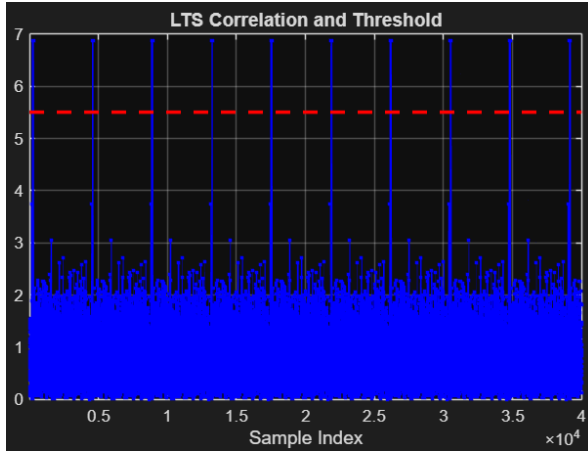
1. rx_pack_detect.m and rx_demodulation.m



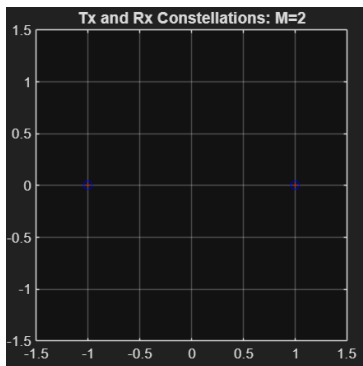
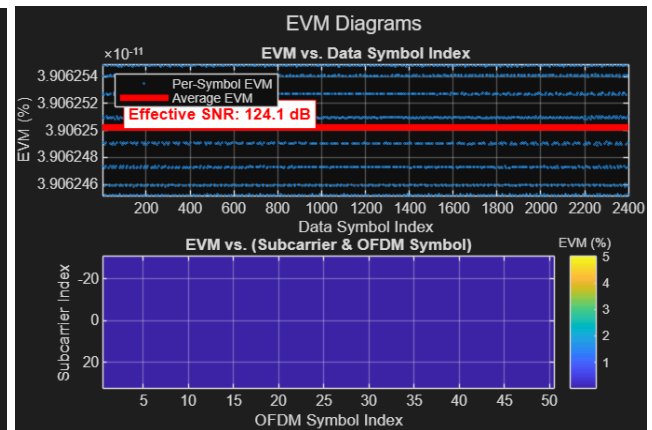
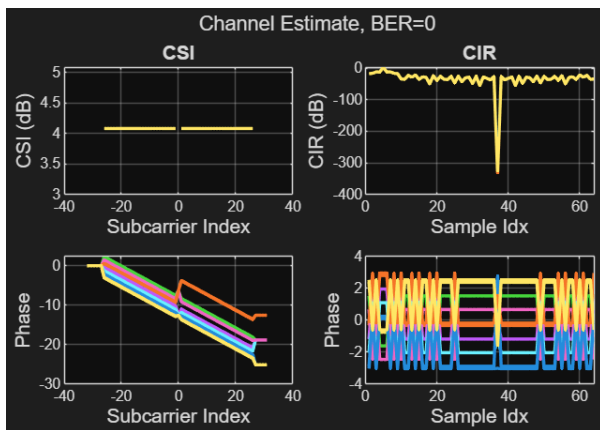
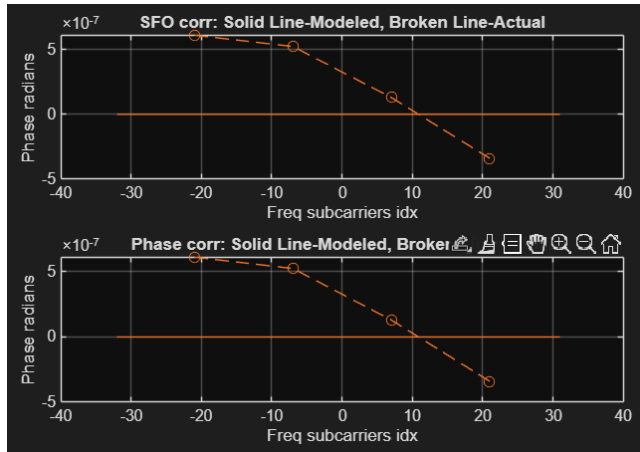
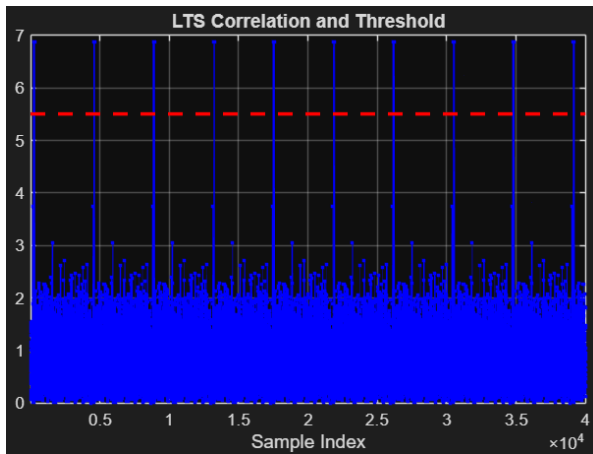
2. rx_channel_estimation.m and rx_equalize_signal.m, CHANNEL_FLAG=1



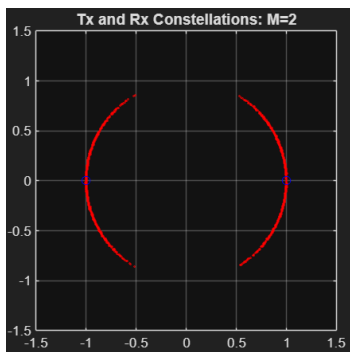
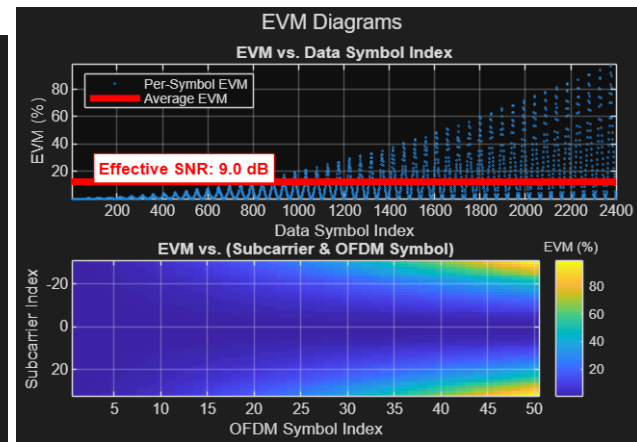
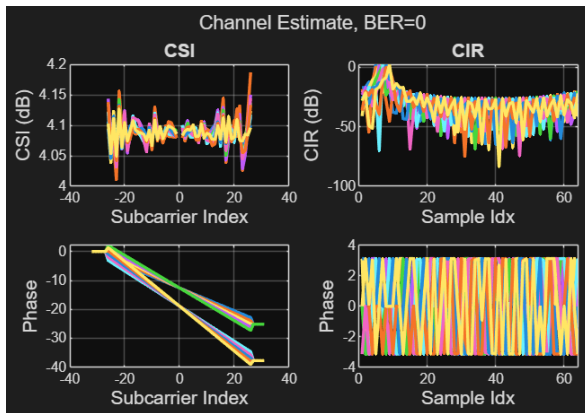
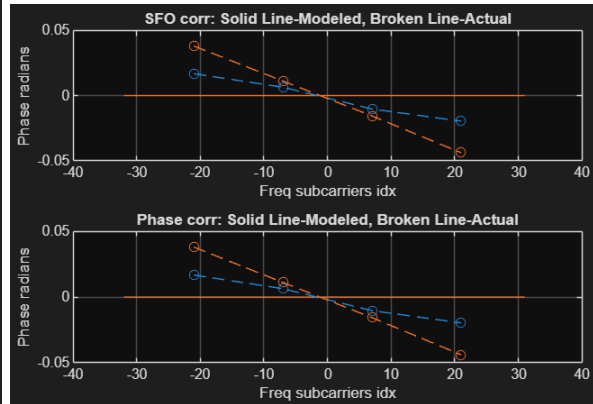
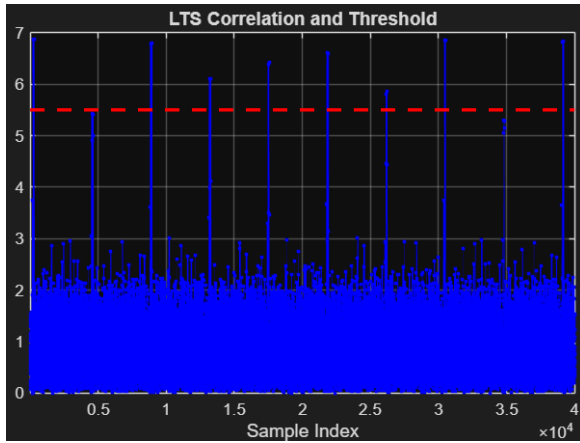
3. rx_cfo_correction.m, CFO_FLAG=1, CFO=100, CHANNEL_FLAG=0
 - a. ofdm_params.DO_APPLY_CFO_CORRECTION=FALSE



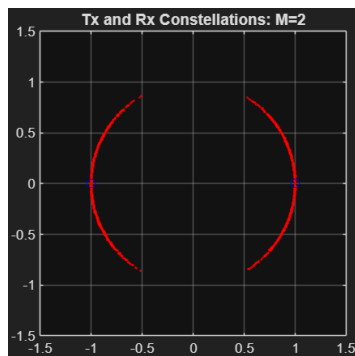
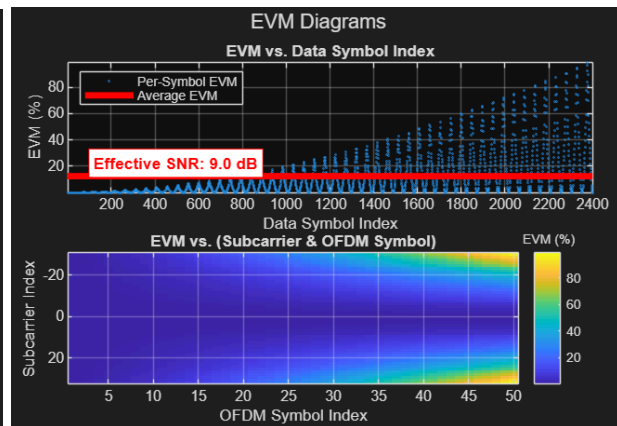
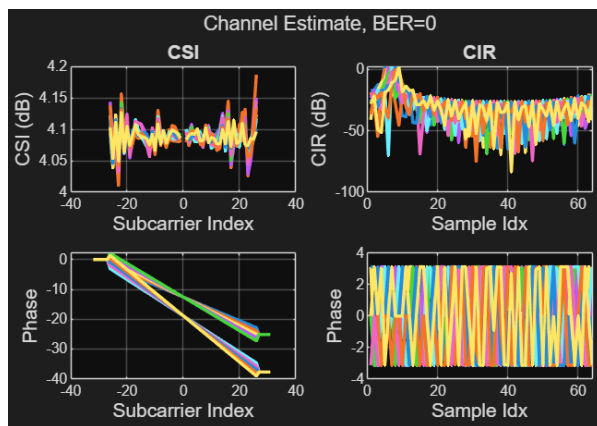
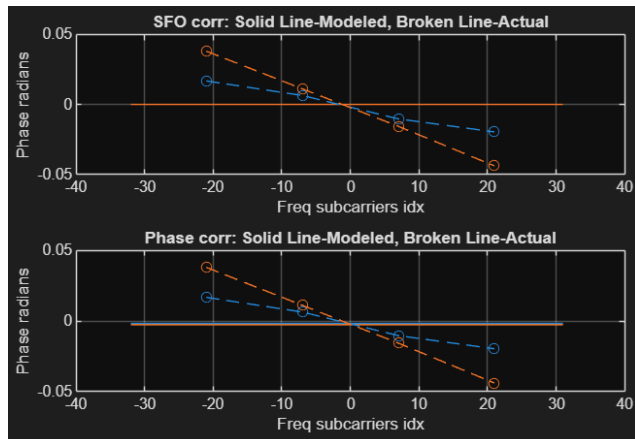
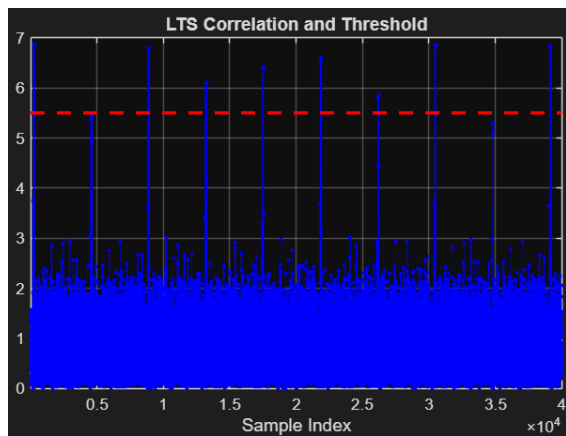
b. ofdm_params.DO_APPLY_CFO_CORRECTION=TRUE



4. rx_sfo_and_phase_error_correction.m, CFO_FLAG=0, SFO_FLAG=1, SFO=100
 - a. ofdm_params.DO_APPLY_SFO_CORRECTION=FALSE
 - i. ofdm_params.DO_APPLY_PHASE_ERROR_CORRECTION=FALSE

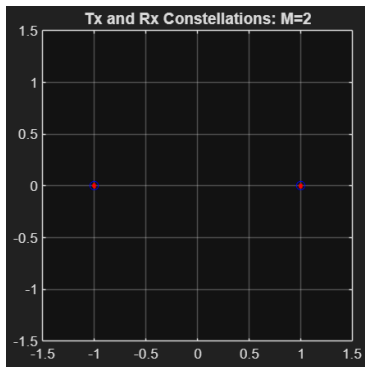
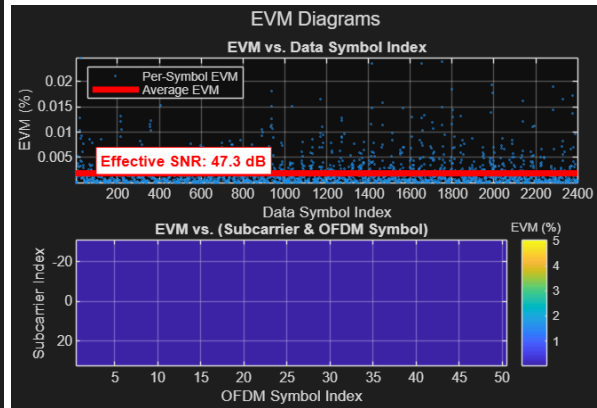
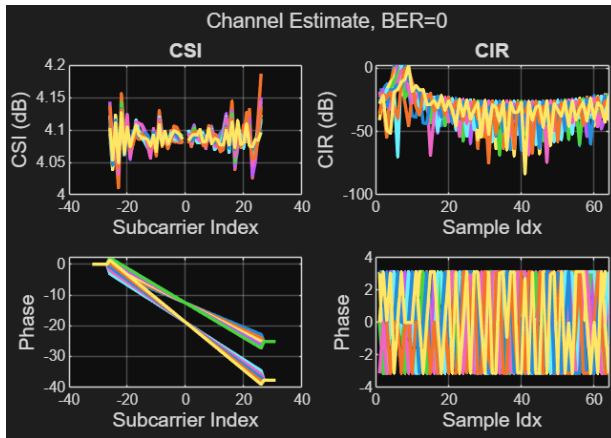
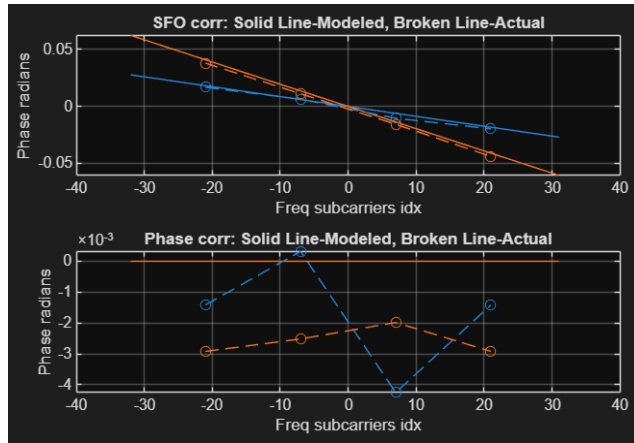
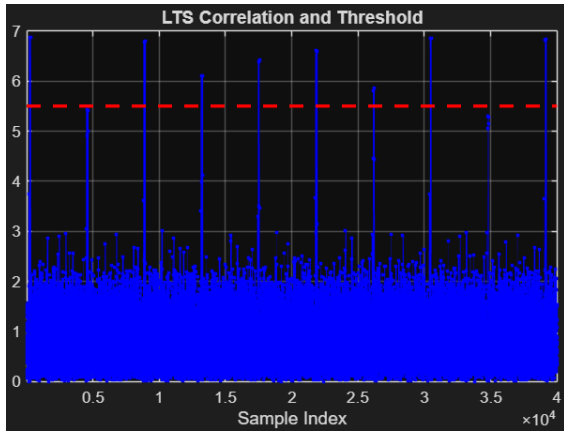


ii. `ofdm_params.DO_APPLY_PHASE_ERROR_CORRECTION=TRUE`

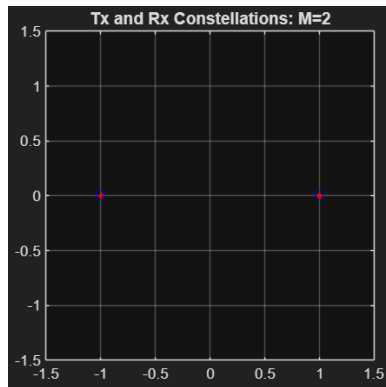
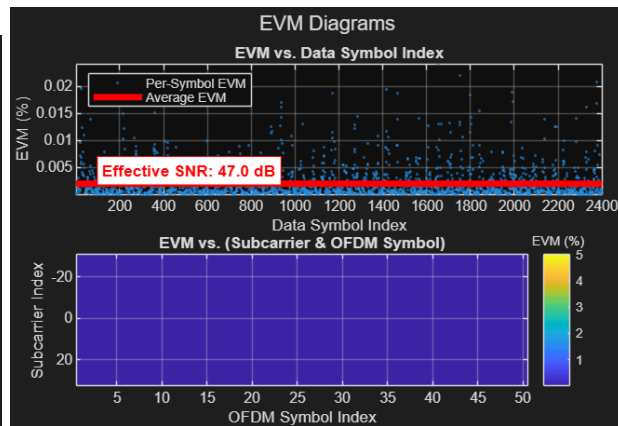
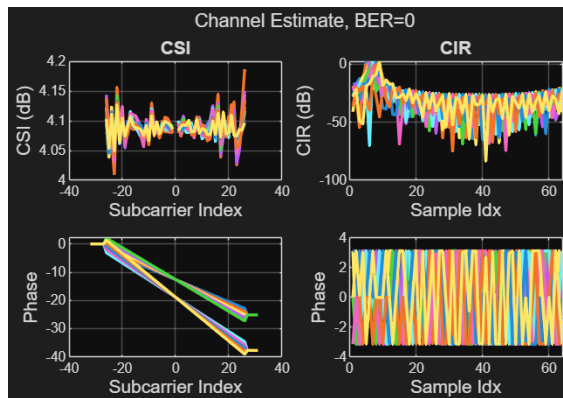
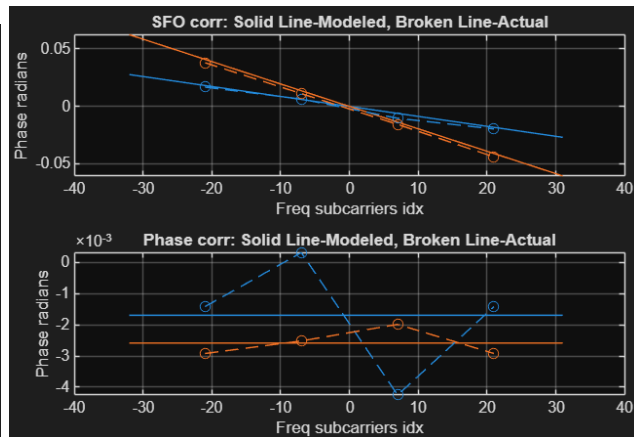
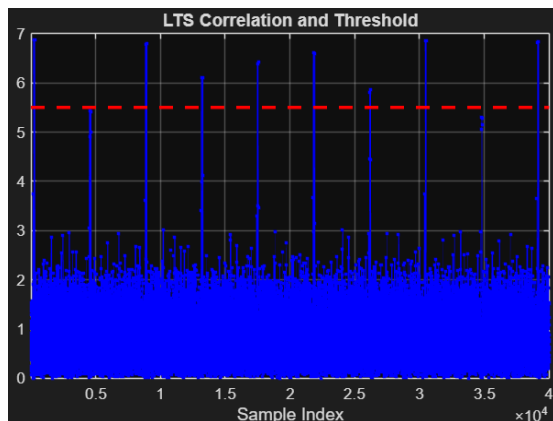


b. ofdm_params.DO_APPLY_SFO_CORRECTION=TRUE

i. ofdm_params.DO_APPLY_PHASE_ERROR_CORRECTION=FALSE

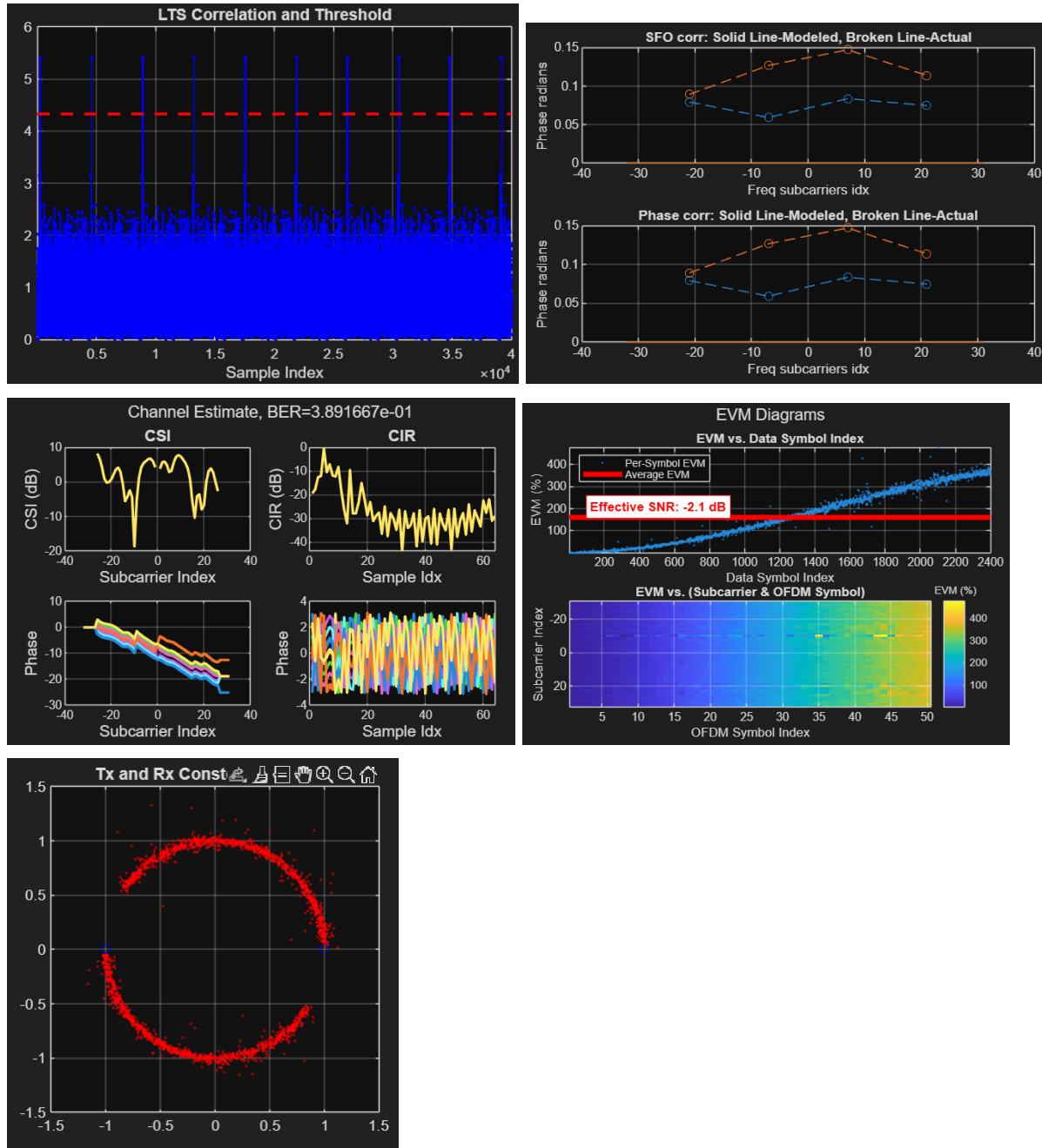


ii. ofdm_params.DO_APPLY_PHASE_ERROR_CORRECTION=TRUE



Deliverables:

1. CFO_FLAG = 1, CHANNEL_FLAG = 1,
ofdm_params.DO_APPLY_CFO_CORRECTION=0

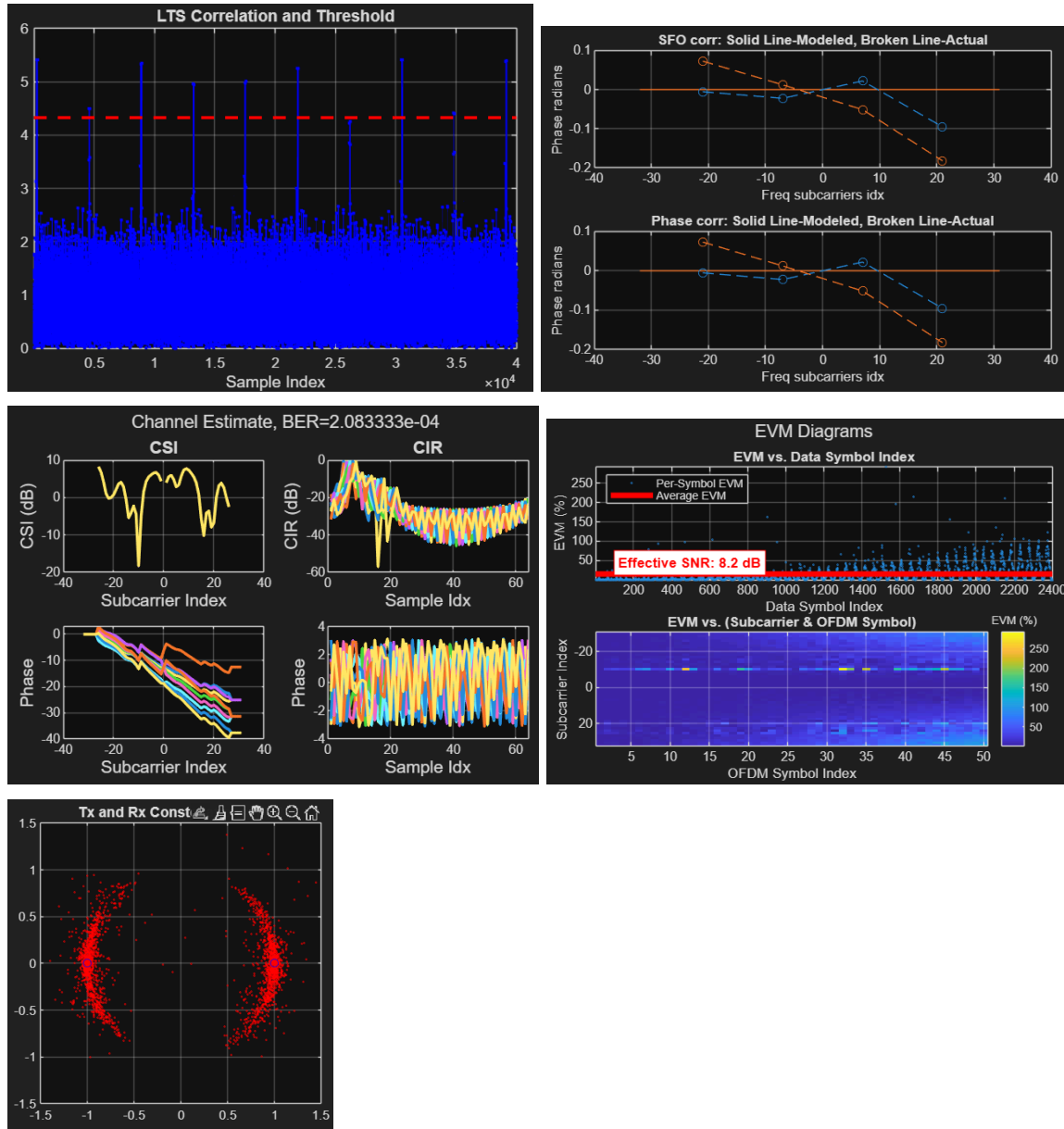


There is a rotation from the expected points at +1 and -1 counter clockwise, almost reaching the opposite point. The EVM vs Data Symbol Index shows an average EVM around 175, while the individual points grow over time from zero to nearly 400. It also shows an effective SNR of -2.1dB. The EVM vs subcarrier & OFDM symbols graph shows the same growth over time, with a line across the subcarrier index of about 10 with sporadically higher EVM. The channel estimate plots show the phase vs subcarrier index with a downward slope, and CIR vs sample idx

with a noisy downward then upward slope. The BER is $3.8e-1$. The shapes of these graphs remain similar across all three tests.

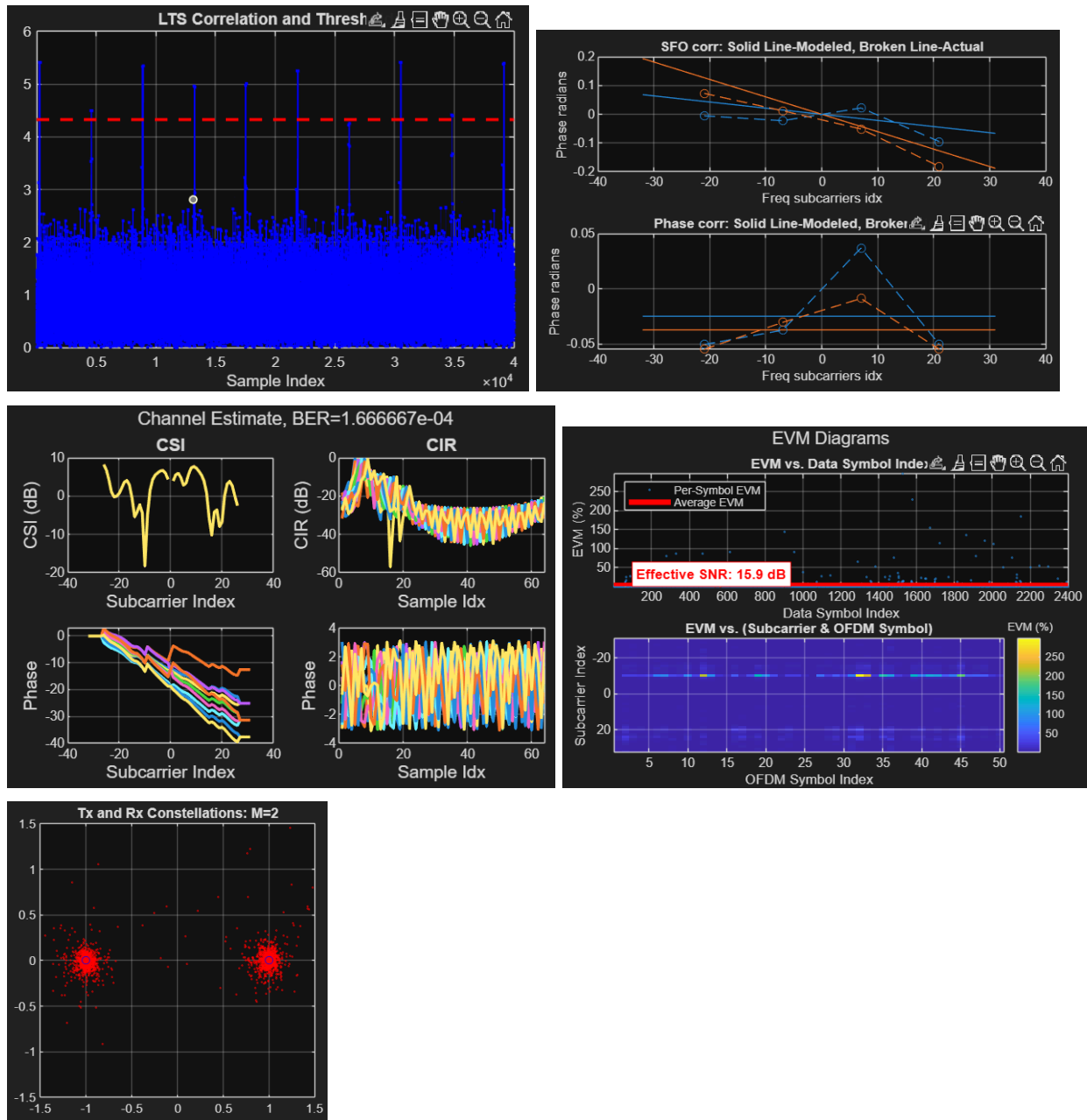
The EVM grows with symbol index because of the accumulating phase error turning small offsets into large symbol errors over the packet, resulting in very poor effective SNR and BER.

2. ofdm_params.DO_APPLY_CFO_CORRECTION=1,SFO_FLAG = 1



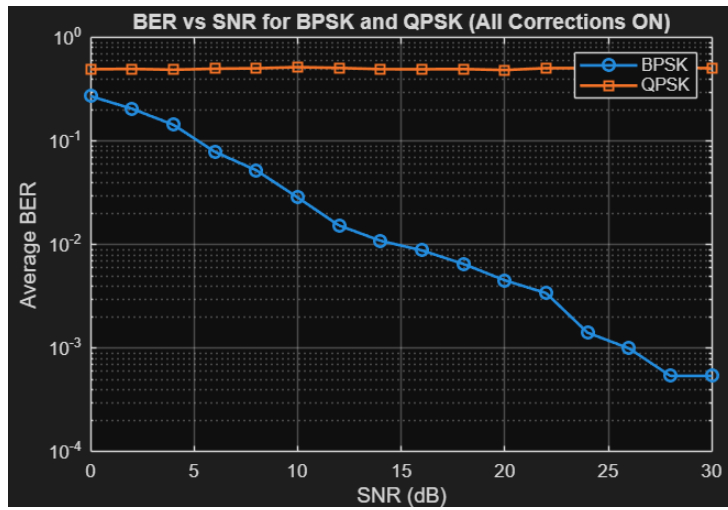
The rotation in the constellation is still there, but it's now "centered", not straying from positive to negative or vice versa. The EMV shows an effective SNR of 8.2dB. The BER is 2.1×10^{-4} . The EVM stays mostly low, especially at the subcarrier index 0, but near the edges it slowly grows as SFO causes per-subcarrier phase curvature. This correction is a vast improvement over part 1.

3. `ofdm_params.DO_APPLY_SFO_CORRECTION=1,`
`ofdm_params.DO_APPLY_PHASE_ERR_CORRECTION=1, params.N_OFDM_SYMS`
`= 1000`



The constellation is now two dots, still noisy, but no phase issues. The EMV is near zero and shows no growth over time, with an effective SNR of 15.9dB The BER is 1.6e-4 This is a huge improvement. The large packets do amplify some of the estimation error as there is no correction throughout the packet, but despite this, it still gives the best results thus far.

4. `params.N_OFDM_SYMS = 50`, `params.N_OFDM_SYMS = 50`, traverse SNRdB,



They both start off quite high at very low SNR, but BPSK quickly improves its BER as SNR grows higher.